Need What and Why: A Comparative Analysis of Behavioral Performance, Social Support of Patients in Online Health Communities

Shugang Li
School of Management
Shanghai University
Shanghai, China
luck_li@shu.edu.cn

Tianyi Jiang*
School of Management
Shanghai University
Shanghai, China
jiangty@shu.edu.cn

Abstract

With the rise of online healthcare, patients increasingly exchange social support through online health communities (OHCs). Different diseases result in varying needs for information and emotional support. This study addresses the gap in understanding these differences by comparing chronic and acute disease patients. We analyzed data from 53,245 chronic disease users and 492,495 posts from CHOHC, and 23,659 acute disease users with 149,095 posts from ACOHC. Using user profiling, pre-trained models, and social network analysis, we found that chronic disease users mainly seek emotional support, while acute disease users prioritize informational support. The findings offer practical insights for designing targeted support for both patient groups.

CCS Concepts

• **Information systems** → Data management systems; Information integration.

Keywords

Online health community, Social support, Social network analysis, Chronic and acute diseases

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1 Introduction

The development of online health communities (OHCs) has facilitated patients and their families in exchanging information and sharing resources, enabling them to seek or provide social support. Existing studies indicate that social support is crucial in alleviating the negative impacts of diseases, particularly chronic conditions, on health[1]. Thus, understanding the behavioral differences and social support needs among various disease groups is vital. OHCs serve as platforms where patients use the internet to share health-related knowledge, playing a significant role in health management and recovery[2]. Social support encompasses not only informational and emotional support but also companionship and substantive support[3]. Research shows that patients sometimes seek only informational support in OHCs, indicating that their social support needs vary depending on disease type, user characteristics, and other factors[4].

This study aims to comparatively analyze the characteristics of users with different disease types and their social support performance from a large volume of postings in online health communities. We categorize disease types into acute and chronic, employing various methods to analyze users' emotions, behaviors, and social support needs, enriching user profiles and enhancing the understanding of social support types in OHCs.

The research questions addressed in this study are:

Q1: What are the overall behavioral characteristics of users with different diseases in online health communities?

Q2: What are the differences in social support performance between users with acute and chronic diseases?

Q3: What are the correlations between user behavioral characteristics and social support needs in online health communities?

By using deep learning models and social network analysis, we explore user behaviors, emotions, and support needs, enriching the understanding of different patient groups and their social support dynamics.

2 literature review

2.1 Acute diseases and chronic diseases

This study categorizes diseases into chronic and acute based on clinical features, pathological course, and pathogenesis[5]. Chronic diseases, such as diabetes and cancer, have a slow onset and long course, contributing to global morbidity and mortality[6]. Acute diseases, including infectious diseases like SARS and COVID-19, progress rapidly with severe symptoms[7]. Treatment for acute diseases often aims for a cure, while chronic conditions focus on management. Effective communication and active listening are crucial for chronic disease patients, whose social support needs differ from those with acute illnesses.

*Corresponding author.

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Table 1: The statistics of the captured data.

Data	CHOHC	ACOHC
Users	53,245 chronic disease users	23,659 acute disease users
Posts	492,495 postings by chronic disease users	149,095 postings by acute disease users

2.2 Social support

Social support theory, introduced by Durkheim, refers to individuals receiving support, information, and feedback. In online health communities, this includes exchanging information, emotions, and companionship. Informational support provides advice, emotional support offers trust, and companionship involves casual, non-medical interactions. Social support needs vary by disease type and psychological state. For instance, rare disease patients require more emotional and informational support, while during COVID-19, neuroinflammatory patients benefited from increased support, and eating disorder communities shifted towards more informational exchanges[8]. More research is needed to examine social support differences in chronic versus acute conditions.

3 Data and Methodologies

3.1 Data sources

In this study, we used web crawlers to collect public data on chronic and acute diseases from representative Chinese online health communities, including Baidu Medical Forum and Sweet Homeland. The chronic disease community (CHOHC) covers cardiovascular disease, cancer, diabetes, and chronic kidney disease, while the acute disease community (ACOHC) includes influenza A, COVID-19, acute appendicitis, and cerebral hemorrhage (Table 1). These platforms were chosen for their large user base, high engagement, and comprehensive disease coverage, facilitating data-rich, comparative analysis.

Data was collected from January 2015 to April 2023, including user and posting data from both CHOHC and ACOHC. For the crawled data, data cleansing was performed to ensure the validity and reliability of the data. Obvious adverts and irrelevant posts were first filtered out to ensure the quality of the data. Secondly, we ensured the activity and authenticity of the selected users by setting minimum interaction thresholds (e.g. only users who participated in posting or replying more than a certain number of times were included in the analysis). Since the data comes from a publicly available online community, it is not possible to fully verify the real identity of the users. However, a certain level of screening was performed based on IP address and frequency of activity to exclude possible machine accounts or automated behavior. For example, in the diabetes forum, posts were categorized by disease type (Type I, Type II, and complications). The original posts and replies were crawled with a post-to-reply ratio of about 1:4. The dataset spans tens of thousands of topics, and the number of chronic disease users and posts was notably larger than that of acute disease users.

3.2 Study 1: User profile

The goal of this study is to analyze the performance of patient users and their social support by first constructing user profiles. User profiling helps understand user needs, offer personalized services, and address the mismatch between user needs and available services in online health communities[9].

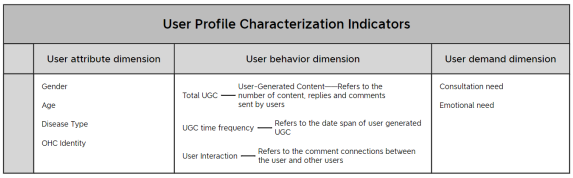

Figure 1: User profile feature system.

The user profiling system in this study consists of three dimensions: user information (e.g., gender, age, disease type), user behavior (e.g., total user-generated content, interaction frequency), and user emotions (positive or negative sentiment). User information is obtained directly from the community, while user behavior is measured by the number of posts, replies, and comments. The UGC time frequency reflects user engagement over time. The specific indicator system is constructed as Figure 1 shown below.

User sentiment is analyzed using the Baidu Sentiment Analysis interface. To cluster users, we apply the K-means algorithm, chosen for its efficiency and scalability, making it suitable for handling large datasets in this study.

3.3 Study 2: Segmentation of social support types

In this section, we categorize user-posted content (posts and comments) to identify the social support needs of patients with different diseases. Existing research has used both manual coding and automated methods to categorize social support. For instance, Wang's study used text mining to automatically classify large-scale data[10].

In this study, we use the ERNIE2.0 machine learning model to categorize user posts into five social support categories: seeking informational support, providing informational support, seeking emotional support, providing emotional support, and accompanying support. Accompanying support refers to non-medical topics such as daily activities and is not split into "seeking" or "providing" categories. Table 2 gives the example of social support for user-posted comments in two communities.

We used Baidu AI's intelligent labeling function for data annotation. The ERNIE2.0 model, similar to BERT, improves the pre-training process by extracting lexical, syntactic, and semantic information from the corpus. First, 3,500 texts were manually annotated through stratified sampling from 550,000 texts, and the ERNIE2.0 model was used to label the remaining data automatically. After several rounds of optimization, an annotation accuracy of 86.3% was achieved. For further validation, Aliyun's iTAG platform was

Table 2: Exemplar Posts of Social Support in Each Category.

Type	CHOHC	ACOHC
Seek information support (SIS)	Do I need to inject insulin in my case?	My child has the flu and has a fever of 38 degrees in the middle of the night, what medicine should I take?
Provide information support (PIS)	All pregnancies are treated with insulin, and the side effects of the medication can have an effect on the child.	Quadrivalent influenza vaccine is one of the preventive vaccines recommended by the WHO, mainly used to prevent the four influenza viruses, quite effective.
Seek emotional support (SES)	Can someone please comfort me? It feels like it's impossible to hold on!	The kids at home have been coughing and it hurts me so much to hear that.
Provide emotional support (PES)	Don't be so pessimistic, eat right, exercise properly and stick with it and you'll be fine!	Don't be discouraged. We'll beat this virus.
Accompany support (COM)	Day 8 of the diabetes diagnosis. Hang on, family.	The number of covid-19 is increasing a lot day by day, so wear a good mask!

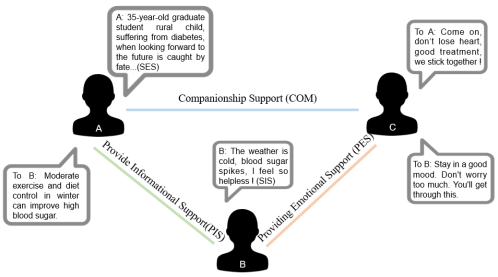

Figure 2: Display of social network relationships between users.

also used, and its accuracy was lower at 78.3%. Therefore, Baidu AI's results were selected for this study.

3.4 Study 3: Social support network analysis

3.4.1 Word co-occurrence network construction. The co-word network analyzes keyword relationships based on their co-occurrence in posts. Text data is processed by calculating word frequencies, merging synonyms, and removing low-frequency words. A co-occurrence matrix is generated, where nodes represent keywords and edges indicate co-occurrence. Larger nodes show more important keywords, and words with many connections are centrally positioned. In this study, all connection lines are of equal thickness for simplicity.

3.4.2 Network analysis of social support. In the online health community, users are represented as nodes in a social network formed through their posts and comments. Types of social support are identified based on user interactions, creating multi-relational connections. For instance, when one user answers another's disease-related question, they are linked by informational support; users sharing diet tips connect through companionship support; and those offering care and comfort are linked by emotional support. The social network relationship between users is shown in Figure 2.

Different types of support form sub-networks that combine into an aggregated network. The number of nodes shows how many users engage in a specific support type, while connection density reflects activity levels. Analyzing this network helps compare user interactions and support performance across different disease communities.

4 Results and Discussion

4.1 User profile results

The K-means algorithm was used to create user profiles for both CHOHC and ACOHC communities. In CHOHC, the optimal clustering occurred with three groups, as shown in Table 3 Category 1 includes newcomers with type I and II diabetes and cancer. These users have low participation, and their posts generally show negative sentiment. Category 2 covers users with type II diabetes, cancer, and chronic kidney disease. They also show moderate engagement, but their posts tend to be negative due to their newcomer status. Category 3 represents long-term users with higher participation and interaction rates. Their posts generally express positive emotions like encouragement and perseverance.

For ACOHC, five distinct categories were identified, as shown in Table 4 Categories 1 and 4 consist mainly of older female patients. Category 4 has higher activity due to concerns related to COVID-19, leading to more frequent posts. Category 2 includes younger users with acute appendicitis, showing low activity, possibly due to the nature of the disease requiring offline surgical treatment. Category 3 is composed of underage patients, with male family members often posting. These users are highly active in online consultations. Category 5 consists of older male patients, primarily suffering from cerebral hemorrhage. Their focus is on medical treatment and consultation.

When comparing the two communities, CHOHC has a larger number of highly engaged users due to the chronic nature of their conditions, which involve long-term interaction and medical advice. In contrast, ACOHC shows lower engagement overall, as users with acute conditions tend to rely more on offline emergency treatment. Consequently, ACOHC has a faster turnover of users and less continuous participation, while CHOHC shows higher stickiness and long-term engagement.

Table 3: User profile results of CHOHC.

Type	Type1	Type2	Type3
Type of disease	I and II diabetes, cancer	II diabetes, cancer, chronic kidney disease	Cardiovascular disease, II diabetes, chronic kidney disease
OHC identity	Newcomer	Newcomer	Resident users
Interaction	Low	Middle	High
Total UGC	Low	High	High
UGC time frequency	Low	High	High
Emotional tendency	Negative	Negative	Positive

Table 4: User profile results of ACOHC.

Type	Type1	Type2	Type3	Type4	Type5
User gender	Woman	Woman	Man	Woman	Man
User age	Elderly patient	Young patient	Minor patient	Elderly patient	Elderly patient
Symptom	Suspected flu	Acute abdominalgia	Acute abdominalgia/flu	Suspected flu/Covid-19	Cerebral hemorrhage
Total UGC	Low	Low	High	High	Low
UGC time frequency	Low	Low	High	High	Low
Demand dimension	Therapeutic consultation	Therapeutic consultation	Therapeutic consultation	Pathological knowledge	Therapeutic consultation

Table 5: Comparison of social support behavior between different diseases.

Social Support type	Results for CHOHC	Results for ACOHC
SIS	17.2%	22.82%
PIS	24.9%	41.00%
SES	16.92%	6.27%
PES	20.63%	9.33%
COM	12.45%	3.50%
Total Posts	492495	149095

4.2 Social support results of users

Table 5 shows distinct differences in social support needs between CHOHC and ACOHC. In CHOHC, users seeking informational (17.2%) and emotional support (16.92%) are nearly equal, with slightly more users providing informational support (24.9%). In contrast, ACOHC is dominated by informational support, with 22.82% seeking and 41% providing it, while emotional support is minimal (seeking 6.27%, providing 9.33%).

CHOHC users are more engaged and dependent on emotional and companionship support, while ACOHC users prioritize quick information exchange with lower emotional engagement. Combined with the conclusions of Studies 1 and 2, we provide insight into the reasons behind different disease needs. Chronic diseases require long-term management, and patients need emotional support to maintain motivation and cope with uncertainty. In contrast, acute diseases such as COVID-19 are short-term and urgent, prompting patients to seek information support for rapid decision making and recovery.

4.3 Results of network analysis

4.3.1 Word co-occurrence network of CHOHC and ACOHC.. This section analyzes the word co-occurrence networks of posts used to categorize social support in CHOHC and ACOHC, as shown in Figure 3. Due to the large number of posts, a random sample was selected from each community, covering users from different clusters. These posts were analyzed to visualize word relationships while preserving semantic links.

In the CHOHC network, keywords like ”cancer” and ”blood sugar” are central, indicating topics related to treatment and emotions. Some areas, like those around ”insomnia,” focus on emotional communication, such as anxiety and helplessness. Thus, CHOHC primarily reflects both informational and emotional support. In ACOHC, keywords like ”influenza” and ”appendix” are prominent, with most topics revolving around treatment, symptoms, and medication. A smaller portion covers emotional topics like pain and fear, indicating a focus on informational support in acute illness communities. The comparison shows that CHOHC emphasizes

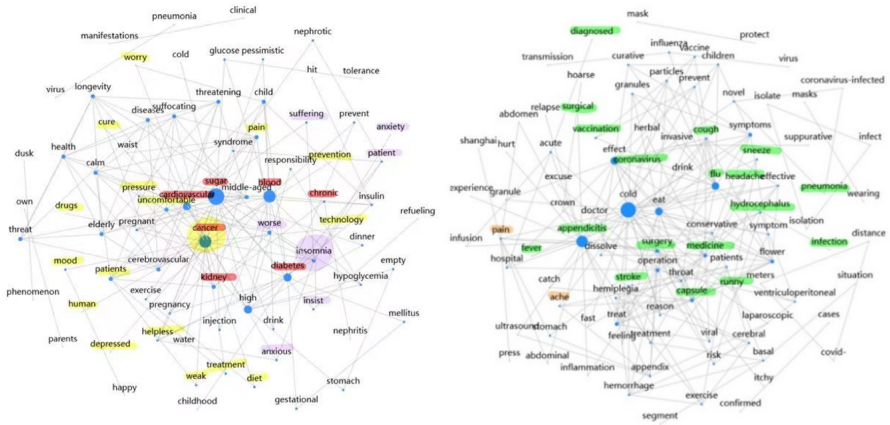

Figure 3: Word co-occurrence network in CHOHC and ACOHC.

Table 6: Results for Descriptive Statistics of CHOHC.

Properties	Aggregatednetwork	Sub-network				
		SIS	PIS	SES	PES	COM
Node	31,258	29,935	19,629	20,591	28,599	19,672
Connection	145,928	128,765	82,153	126,542	122,095	93,671
Density	0.05	0.04	0.03	0.05	0.05	0.04

Table 7: Results for Descriptive Statistics of ACOHC.

Properties	Aggregatednetwork	Sub-network				
		SIS	PIS	SES	PES	COM
Node	16,927	13,663	13,258	10,801	10,263	9538
Connection	56,283	42,362	38,965	18,334	15,956	13,505
Density	0.03	0.03	0.02	0.004	0.003	0.001

both informational and emotional support, while ACOHC focuses more on treatment-related information. This analysis aligns with the earlier findings on social support segmentation.

4.3.2 Social network of CHOHC and ACOHC.. This study analyzes multiple social networks in CHOHC and ACOHC. In CHOHC, as shown in Table 6, the Seeking Information Support (SIS) and Providing Emotional Support (PES) sub-networks have the most nodes and similar numbers of edges, indicating active users for both informational and emotional support. The Seeking Emotional Support (SES) sub-network is more densely connected than Providing Information Support (PIS), showing stronger connections among emotional support users.

In ACOHC, as shown in Table 7, the largest sub-networks are Seeking Information Support (SIS) and Providing Information Support (PIS), where users are most active and post frequently. Overall, user activity and connections are stronger in CHOHC, with more balanced participation across sub-networks, while ACOHC shows high activity mainly in informational support, with other sub-networks being more sparse.

5 Conclusions

Online health communities play a vital role in providing medical advice and social support, especially in the wake of covid-19. This study found that chronic illness users (CHOHCs) sought both emotional and informational support, while acute illness users (ACOHCs) prioritized timely, accurate information for treatment. Healthcare providers should provide emotional counseling for chronic patients and ensure accurate content for acute patients. At the same time, the online platform should be differentiated to meet the different needs of different patients, which can improve the quality of care. Policymakers can integrate targeted support systems into public health strategies and encourage collaboration between online communities and health care providers to ensure effective information and emotional support delivery.

Due to resource and time constraints, we focus on specific chronic diseases (e.g. diabetes, cancer) and acute diseases (e.g. influenza A, COVID-19). However, we recognize that the generality may be limited and suggest that this is a direction for future research. Future research could explore more diseases and more

types of online health communities to validate and expand on our findings.

Acknowledgments

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